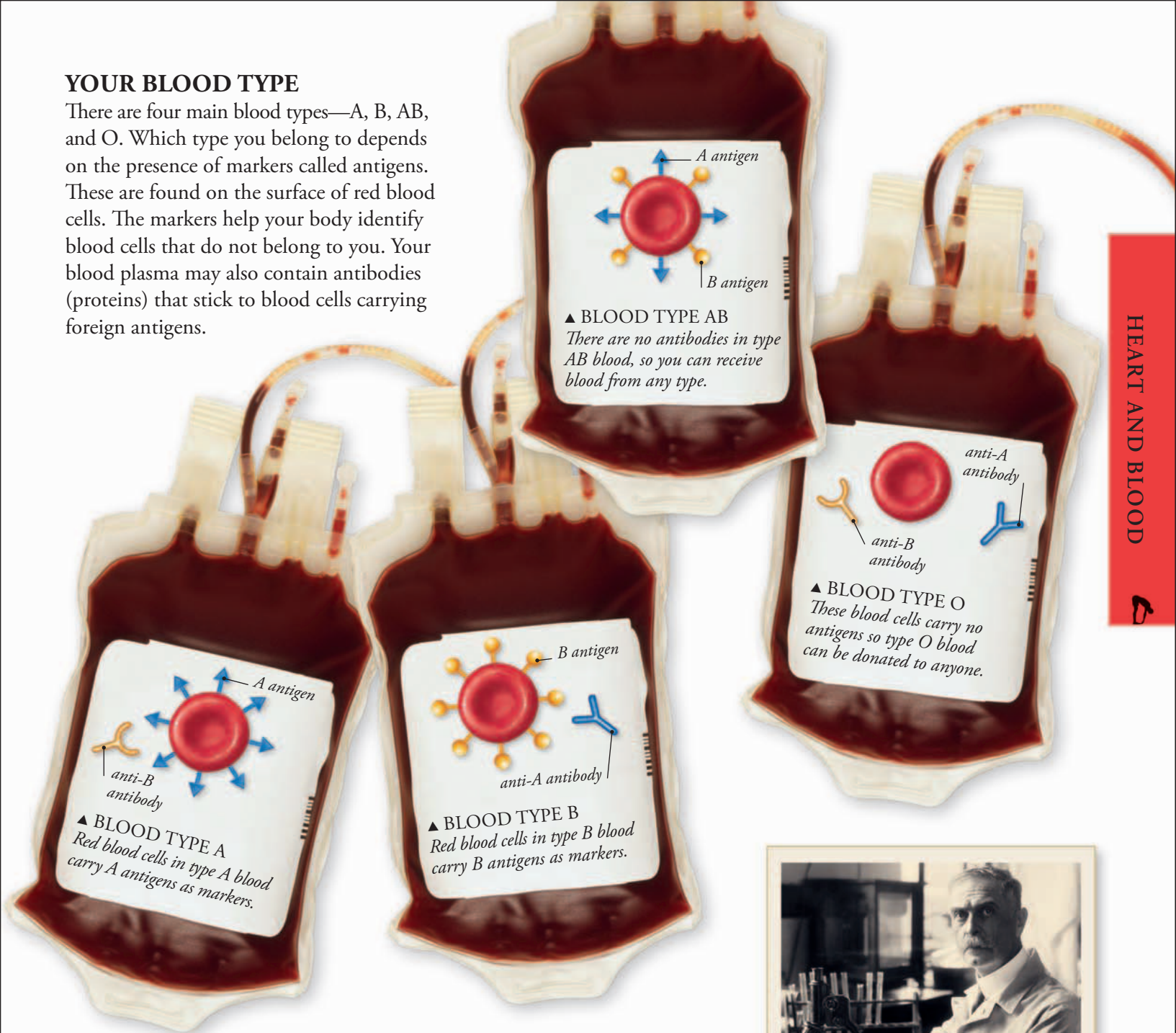


YOUR BLOOD TYPE

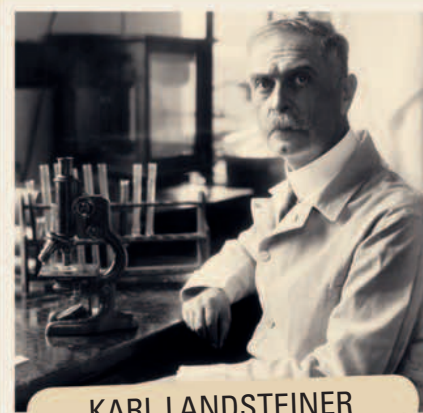
There are four main blood types—A, B, AB, and O. Which type you belong to depends on the presence of markers called antigens. These are found on the surface of red blood cells. The markers help your body identify blood cells that do not belong to you. Your blood plasma may also contain antibodies (proteins) that stick to blood cells carrying foreign antigens.



▲ **BLOOD TRANSFUSION** Blood can be donated to a patient who needs blood. However, a transfusion of the wrong blood type can make the blood cells clump together and block the blood vessels, so it is important to find the correct match.

CHANGING COLOR

Your blood's color alters as it travels around your body. This is because the hemoglobin inside red blood cells changes color as it picks up or releases oxygen. After red blood cells have collected oxygen in your lungs, the blood flowing along arteries is bright red. After red blood cells have released oxygen in the tissues, blood traveling along veins is a dark purple-red.



KARL LANDSTEINER

Until the early 1900s, very little was known about why blood transfusions failed. The breakthrough came in 1901, when Austrian scientist Karl Landsteiner experimented on himself and his team and discovered the four main blood types. Blood could now be transfused safely from a donor to a recipient who shared the same blood type, saving countless of lives.